Singular Value Decomposition
& more

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Preliminaries on Matrices

- The complex vector space $\mathbb{C}^n$: Space that contains all the vectors $(z_1, z_2, \dots, z_n)$ of complex entries. (Consider these as column vectors.)
- $(z_1, z_2, \dots, z_n) + (w_1, w_2, \dots, w_n) = (z_1 + w_1, z_2 + w_2, \dots, z_n + w_n)$.
- $a(z_1, z_2, \dots, z_n) = (az_1, az_2, \dots, az_n)$ $(a \in \mathbb{C})$

- The complex dot product $\cdot$ on $\mathbb{C}^n$:
  $\boldsymbol{x} \cdot \boldsymbol{y} = (\sum_{i=1}^n x_i\, \boldsymbol{e}_i) \cdot (\sum_{i=1}^n y_i\, \boldsymbol{e}_i) = \sum_{i=1}^n x_i\, \overline{y_i}$.

Ex) $\boldsymbol{x} \cdot \boldsymbol{x} = \sum_{i=1}^n x_i\, \overline{x_i} = |x_1|^2 + |x_2|^2 + \cdots + |x_n|^2 \geq 0$.

$\boldsymbol{a} \cdot (\boldsymbol{b} + \boldsymbol{c}) = \boldsymbol{a} \cdot \boldsymbol{b} + \boldsymbol{a} \cdot \boldsymbol{c}$, $(\boldsymbol{a} + \boldsymbol{b}) \cdot \boldsymbol{c} = \boldsymbol{a} \cdot \boldsymbol{c} + \boldsymbol{b} \cdot \boldsymbol{c}$, $\quad \boldsymbol{b} \cdot \boldsymbol{a} = \overline{\boldsymbol{a} \cdot \boldsymbol{b}}$.

Preliminaries on Matrices

- Orthogonality : Two vectors $\boldsymbol{v}, \boldsymbol{w}$ are called orthogonal when $\boldsymbol{v} \cdot \boldsymbol{w} = 0$.
- Unitarity : $\boldsymbol{v}$ is a unit vector if $\boldsymbol{v} \cdot \boldsymbol{v} = 1$.

❖ A set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ is an <u>orthogonal basis</u> for $\mathbb{C}^n$ when

Each $\boldsymbol{v}_i$'s are nonzero and $\forall\ 1 \leq i < j \leq n :\ \boldsymbol{v}_i \cdot \boldsymbol{v}_j = 0$.

(This definition implies linear independency)

❖ Moreover, if $\forall i$: $\boldsymbol{v}_i \cdot \boldsymbol{v}_i = 1$, the set becomes an <u>orthonormal basis</u>.

Preliminaries on Matrices

- M's conjugate transpose M^* is the matrix whose coordinates are $[M^*]_{ij} = \overline{M_{ji}}$. If $H = H^*$, then H is called Hermitian.
- An $n \times n$ matrix M is unitary if (and only if) the columns of M constitute an orthonormal basis for $\mathbb{C}^n$. Here, M is invertible, $M^{-1} = M^*$, and it is also unitary. If Q_1, Q_2 are unitary, then $Q_1 Q_2$ also is.

❖ Theorem. When A is Hermitian, $\exists D(\text{real diagonal}),\ \mathrm{Q}(\text{Unitary})$: $\mathrm{A} = \mathrm{QDQ}^{-1}$. Moreover, if additionally A is real(i. e. symmetric), then Q can be taken real. (Real unitary matrices are called orthogonal.)

Preliminaries on Matrices

Ex) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is symmetric; $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^{-1}$.

The matrix is <u>orthogonally diagonalizable</u>.

Ex) $\begin{bmatrix} 1 & 1+2i \\ 1-2i & -3 \end{bmatrix}$ is Hermitian, as $\begin{bmatrix} 1 & 1+2i \\ 1-2i & -3 \end{bmatrix}^* = \begin{bmatrix} 1 & 1+2i \\ 1-2i & -3 \end{bmatrix}$.

$\begin{bmatrix} 1 & 1+2i \\ 1-2i & -3 \end{bmatrix} = Q \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix} Q^{-1}$ where $Q = \begin{bmatrix} \frac{-1-2i}{\sqrt{30}} & \frac{1+2i}{\sqrt{6}} \\ 5/\sqrt{30} & 1/\sqrt{6} \end{bmatrix}$.

The matrix is <u>unitarily diagonalizable</u>.

The Singular Value Decomposition(SVD)

① Square matrices

- For any $n \times n$ complex matrix A, consider an other $n \times n$ matrix A^*A.

- $(A^*A)^* = A^*(A^*)^* = A^*A$ holds; A^*A is Hermitian.

- By the preceding lemma, $A^*A = VDV^{-1}$(D real diagonal, V unitary).

- Let $\mathrm{V} = \begin{bmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \\ | & | & & | \end{bmatrix}$, $D = \begin{bmatrix} \lambda_1 & & O \\ & \ddots & \\ O & & \lambda_n \end{bmatrix}$.

  Notice that $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ is an orthonormal basis for $\mathbb{C}^n$.

The Singular Value Decomposition(SVD)

- $A^*A = VDV^{-1} \Longrightarrow (A^*A)V = VD$. Comparisons with each column gives $\forall i$: $(A^*A)\boldsymbol{v}_i = V[0 \quad \cdots \quad \lambda_i \quad \cdots \quad 0]^T = \lambda_i \boldsymbol{v}_i$. What does it mean?

- This means $\boldsymbol{v}_i \overset{x \to Ax}{\longmapsto} \boldsymbol{w}_i \overset{x \to A^*x}{\longmapsto} \lambda_i \boldsymbol{v}_i$. (Take these $\boldsymbol{w}_i$'s.)

- "When we multiply a vector $\sum_{i=1}^{n} c_i \boldsymbol{v}_i$ by A, we get $\sum_{i=1}^{n} c_i \boldsymbol{w}_i$."

- We shall take $U = \begin{bmatrix} | & | & & | \\ \frac{\boldsymbol{w}_1}{\sqrt{\lambda_1}} & \frac{\boldsymbol{w}_2}{\sqrt{\lambda_2}} & \cdots & \frac{\boldsymbol{w}_n}{\sqrt{\lambda_n}} \\ | & | & & | \end{bmatrix}$. (Hey, but are λ_n's nonnegative?)

The Singular Value Decomposition(SVD)

- Indeed! $\lambda_i = \lambda_i \boldsymbol{v}_i \cdot \boldsymbol{v}_i = A^* A \boldsymbol{v}_i \cdot \boldsymbol{v}_i = \boldsymbol{v}_i^* A^* A \boldsymbol{v}_i = A\boldsymbol{v}_i \cdot A\boldsymbol{v}_i \geq 0.$
- Moreover, $\frac{\boldsymbol{w}_a}{\sqrt{\lambda_a}} \cdot \frac{\boldsymbol{w}_b}{\sqrt{\lambda_b}} = \frac{1}{\sqrt{\lambda_a \lambda_b}}(Av_a \cdot Av_b) = \frac{1}{\sqrt{\lambda_a \lambda_b}}(A^* A v_a \cdot v_b) = \sqrt{\frac{\lambda_a}{\lambda_b}}(v_a \cdot v_b) = \begin{cases} 0 & (a \neq b) \\ 1 & (a = b) \end{cases}.$
- Hence U becomes unitary!
- "When we multiply a vector $\sum_{i=1}^{n} c_i \boldsymbol{v}_i$ by A, we get $\sum_{i=1}^{n} c_i \boldsymbol{w}_i$."

$$\therefore A(V[c_1 \quad \cdots \quad c_n]^T) = \begin{bmatrix} | & | & & | \\ \boldsymbol{w}_1 & \boldsymbol{w}_2 & \cdots & \boldsymbol{w}_n \\ | & | & & | \end{bmatrix} [c_1 \quad \cdots \quad c_n]^T = U \begin{bmatrix} \sqrt{\lambda_1} & & O \\ & \ddots & \\ O & & \sqrt{\lambda_n} \end{bmatrix} [c_1 \quad \cdots \quad c_n]^T;$$

$AV = U\Sigma$, $A = U\Sigma V^{-1}$ where $\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & & O \\ & \ddots & \\ O & & \sqrt{\lambda_n} \end{bmatrix}$.

The Singular Value Decomposition(SVD)

② General matrices: This method of finding U and Σ works not only on the square matrices, but also on general matrices.

$$\boldsymbol{v}_i \xmapsto{x \to Ax} \boldsymbol{w}_i \xmapsto{x \to A^* x} \lambda_i \boldsymbol{v}_i$$

- This is the key of generalization; for $m \times n$ matrices s.t. $m > n$, $\boldsymbol{w}_i$'s constitute a *part* of some orthonormal basis for $\mathbb{C}^m$. Extend the *partial* basis $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_n\}$ into $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_m\}$ in order to form U.
- When $m < n$, $\boldsymbol{w}_{m+1}, \dots, \boldsymbol{w}_n$ vanishes; $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_m\}$ becomes an orthonormal basis for $\mathbb{C}^m$. Form U with these vectors as columns.

The Singular Value Decomposition(SVD)

Conclusion.

- Any matrix $A_{m\times n}$ can be expressed as $A = U\Sigma V^{-1}$, where $U_{m\times m}$ and $V_{n\times n}$ are unitary matrices, and $\Sigma_{\mathrm{m}\times n}$ is a matrix that all the nonzero elements belong to the main diagonal and are nonnegative.

$$\Sigma_{11}, \Sigma_{22}, \dots, \Sigma_{\min\{m,n\}\,\min\{m,n\}}$$

- Diagonal entries of Σ = "Singular Values".

The Singular Value Decomposition(SVD)

Ex)

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}^{-1}.$$

$$\begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \xrightarrow{A} \underbrace{\sqrt{3}}_{\text{singular value}} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \xrightarrow{A^*} 3 \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}.$$

An application of SVD

- When A is a square matrix, $A = U\Sigma V$ for unitary matrices U, V and a diagonal matrix Σ.
- It follows that $A = (U\Sigma U^{-1})(UV)$, where $U\Sigma U^{-1}$ is a Hermitian matrix (positive semidefinite) and UV is a unitary matrix.

- Polar Decomposition - A can be expressed as $\mathrm{A} = PU$ where P is positive semidefinite and U is unitary.
- Meaning : (Linear transformation) = (A scaling) X (A rotation).

Well-posed problems

- A problem, which is described by some parameters, is given.
  Here we find the parameters by measuring – which means various unavoidable errors emerge.

- What do we mean by 'parametrized problems'?

- Finding roots of a polynomial, the parameters are the coefficients
- Solving an Initial Value Problem(DE), the parameter is the initial value
- Finding the final location of a golf ball after putting it on the green, the parameters are the strength & the angle of the putting

Well-posed problems

- Well-posed problems : The solution of a well-posed problem changes continuously as the parameter of the problem changes.

- Finding roots of a polynomial is a well-posed problem.
  $x^3 - 6x^2 + 11x - 6$ has three real roots $x = 1, x = 2, x = 3$.
  When the coefficients undergo slight changes, say, they change into
  $x^3 - (6.0024)x^2 + (10.9992)x - (6.0017)$,
  the roots(solution) are adjusted, but only by small amounts;
  $x \approx 1.00247, x \approx 1.98723, x \approx 3.0127$.

Well-posed problems

- Well-posed problems : The solution of a well-posed problem changes continuously as the parameter of the problem changes.

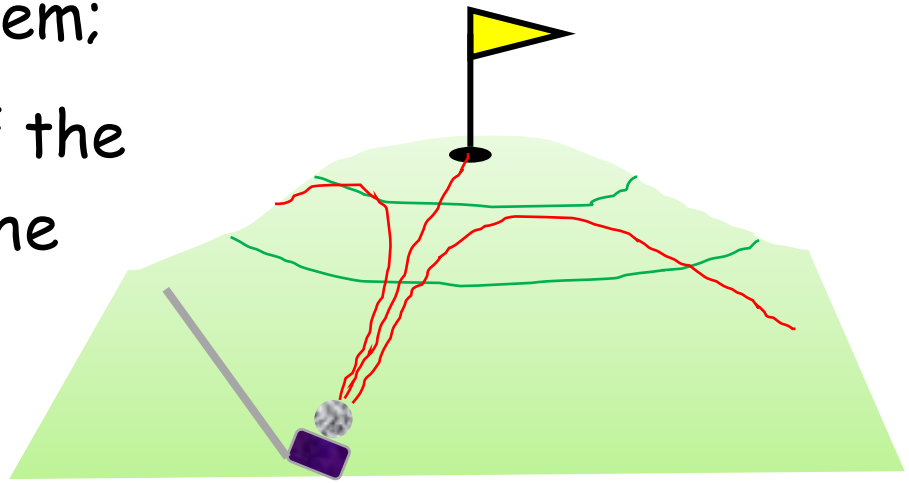
- An example of ill-posed problem (not well-posed) :
  Given a decent Random Number Generator, that generates a real number between 0 and 1, and the number changes by time,
  The problem is to find the number generated at the time t.

0.186329644	0.767102485	0.843142472	0.409926856	0.457526316	0.536870912
$t = 0.2000s$	$t = 0.2001s$	$t = 0.2002s$	$t = 0.2003s$	$t = 0.2004s$	$t = 0.2005s$

Well-posed problems

- However, it is not enough for a problem to be well-posed for it to be solved. Consider the golf ball putting problem;
- As a physical problem, the final location of the ball must have a continuous relation with the initial strength and angle of the putting.

- Which means it is a well-posed problem.



- But small error of parameters can cause a massive difference in the solution (without breaking continuity). In spite of forming a well-posed problem, it is still very difficult to decide how to put a ball.

*Forecasting weather with the current state of weather is a well-posed ill-conditioned problem.

Well-conditioned problems

- Let a problem be a function f, having the parameters x as input and returning the solution $f(x)$. Assume that the solutions lie in a certain metric space(we need to know how 'close' are the solutions).

- The Relative Condition Number $\kappa_x(f)$ of the problem f is defined by

$$\kappa_x(f) = \lim_{\epsilon \to 0} \sup_{\Delta x \leq \epsilon} \frac{|\Delta f(x)| \,/\, |f(x)|}{|\Delta x| \,/\, |x|}.$$

- This expression represents the largest ratio of (size of the error of solution) to (size of the error of parameters) as the parameter error goes small towards 0. (It is dependent of x.)

Well-conditioned problems

Ex) Consider an IVP $y' = y(2-y),\ y(0) = x \in (0,2),\ t \in [0,+\infty)$.

For two real functions $f, g\colon [0,+\infty) \to \mathbb{R}$, define their distances as the 'maximal distance' $|f-g| = \sup_{t\in[0,+\infty)} |f(t) - g(t)|$.

- The solution is $y_x(t) = 2 - \frac{2(2-x)}{xe^{2t}-x+2}$. Hence $|y_{x+\Delta x} - y_x| =$

$$\sup_{t\in[0,+\infty)} |y_{x+\Delta x}(t) - y_x(t)| \approx \begin{cases} \dfrac{\Delta x}{x(2-x)} & (x < 1) \\ \Delta x & (x \geq 1) \end{cases}.$$

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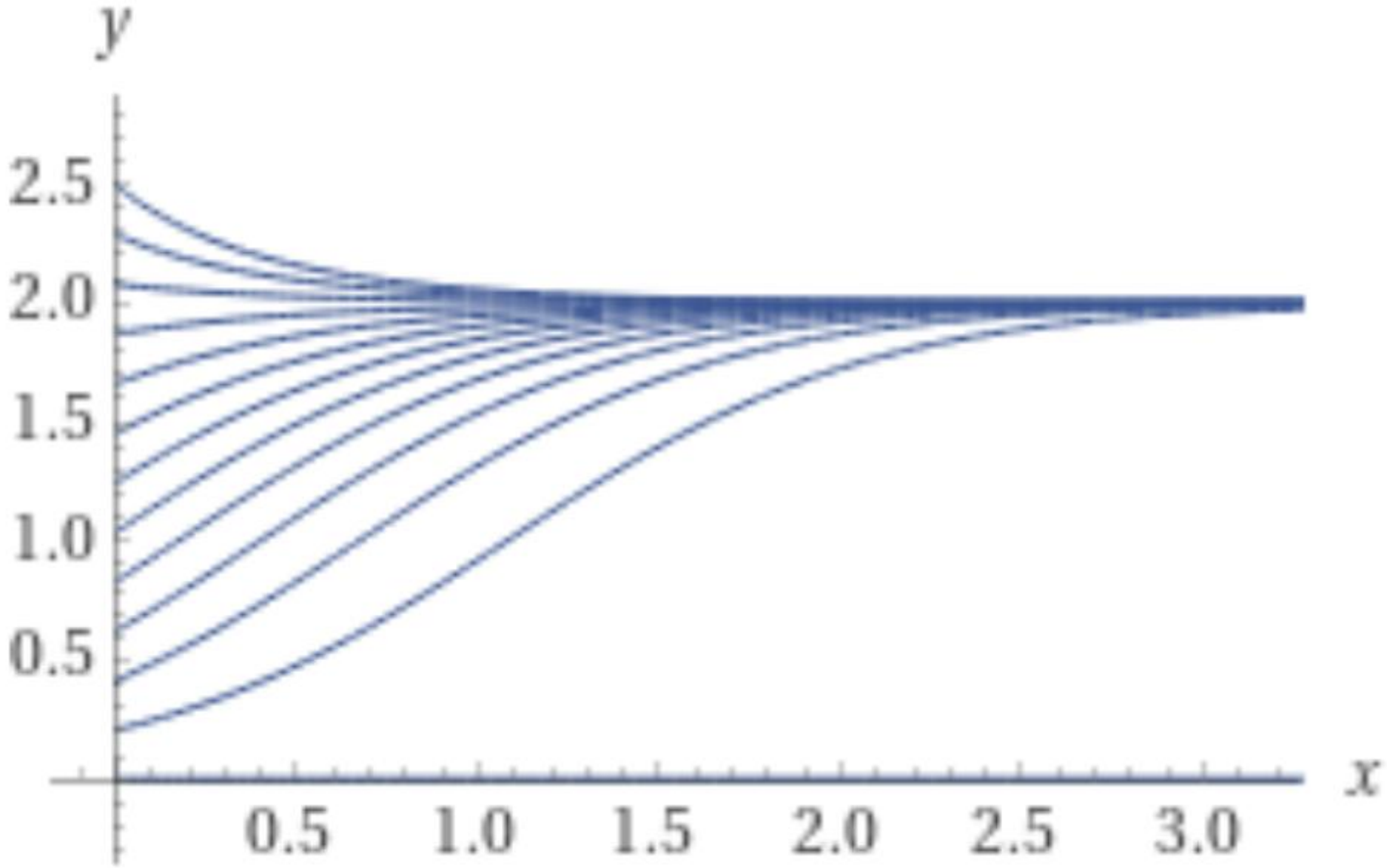

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Condition number of a matrix

- The problem we are interested in :
  Solving the linear system $Ax = b$ (A square, invertible), parameter : b.

- Denote the 'problem function' as f_A; we get $f_A(b) = A^{-1}b$.

❖ The condition number $\mathrm{cond}(A)$ of the matrix A is defined as the maximal relative condition number $\sup_b \kappa_b(f_A)$.

- The smaller the condition number of A is, the smaller the ratio of (the error of the solution of the system) to (the error of b).

Condition number of a matrix

- By the formula, $\mathrm{cond}(A) = \sup_{\boldsymbol{b},\,\boldsymbol{r}} \frac{|A^{-1}(\boldsymbol{b}+\boldsymbol{r})-A^{-1}\boldsymbol{b}|/|A^{-1}\boldsymbol{b}|}{|\boldsymbol{r}|/|\boldsymbol{b}|} = \sup_{\boldsymbol{b},\,\boldsymbol{r}} \frac{|A^{-1}\boldsymbol{r}|/|\boldsymbol{r}|}{|A^{-1}\boldsymbol{b}|/|\boldsymbol{b}|}$.

- To evaluate this expression, let us form the SVD of A;

$$A = U \begin{bmatrix} \sigma_1 & & O \\ & \ddots & \\ O & & \sigma_n \end{bmatrix} V^{-1},\ U,\ V \text{ unitary},\ \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0 \text{ (W.L.O.G.)}$$

Note. If Q is unitary, $|Qx| = (Qx \cdot Qx)^{1/2} = (Q^*Qx \cdot x)^{1/2} = (x \cdot x)^{1/2} = |x|$.

Condition number of a matrix

- By the formula, $\mathrm{cond}(A) = \sup_{\boldsymbol{b},\boldsymbol{r}} \frac{|A^{-1}(\boldsymbol{b}+\boldsymbol{r})-A^{-1}\boldsymbol{b}|/|A^{-1}\boldsymbol{b}|}{|\boldsymbol{r}|/|\boldsymbol{b}|} = \sup_{\boldsymbol{b},\boldsymbol{r}} \frac{|A^{-1}\boldsymbol{r}|/|\boldsymbol{r}|}{|A^{-1}\boldsymbol{b}|/|\boldsymbol{b}|}$.

- $\frac{|A^{-1}x|}{|x|} = \frac{1}{|x|}\left|V\begin{bmatrix} 1/\sigma_1 & & O \\ & \ddots & \\ O & & 1/\sigma_n \end{bmatrix} U^{-1}x\right|$; it has min $\frac{1}{\sigma_1}$, max $\frac{1}{\sigma_n}$! (Why?)

- Therefore, $\mathrm{cond}(A) = \frac{\sigma_1}{\sigma_n}$.

Exercises

- Evaluate $\mathrm{cond}(I_{4\times 4})$ and $\mathrm{cond}\left(\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}\right)$.

  Give explanations for the results.

- If one uses the same formula for the condition number, what would be the condition number of singular(not invertible) square matrices?
  Is it reasonable to define that value as their condition numbers?